Aim: To design a basic Flutter application layout using layout widgets such as Row, Column, Container, Expanded, and Padding.

Theory:

Q1. What is Layout?

Arrangement of widgets on screen using structured positioning rules.

- Controls alignment, spacing, and sizing of child widgets.
- Organizes UI elements into rows, columns, or stacks.
- Ensures responsive design across different screen sizes.
- Managed using layout widgets like Row and Column.

Q2. Why Layout Widgets are needed?

Widgets used to control positioning and arrangement of UI elements.

- Help create structured and readable user interfaces.
- Provide alignment and spacing between multiple widgets.
- Support responsive adjustments for various device screens.
- Allow flexible sizing using constraints and expansion rules.

Q3. Why Container is important?

A versatile widget used for styling and positioning elements.

- Allows adding padding, margin, and alignment properties.
- Supports decoration like color, border, and shadows.
- Can control width and height constraints easily.
- Combines multiple layout and styling features together.

Q4. What is a Padding widget?

Widget that adds empty space around its child widget.

- Creates spacing between child and surrounding widgets.
- Improves visual clarity and layout readability.
- Uses EdgeInsets to define specific spacing values.
- Does not affect child's internal styling properties.

Q5. What is an Expanded widget?

Widget that expands a child to fill available space.

- Used inside Row, Column, or Flex widgets.
- Divides remaining space among multiple children.

- Supports flexible layout design for responsiveness.
- Uses flex property to control space distribution.

Q6. What is Scaffold and why is it needed?

Basic page structure widget providing standard app layout framework.

- Provides app bar, body, drawer, and floating button.
- Implements material design visual layout structure.
- Acts as base structure for every screen.
- Manages snackbars and bottom navigation bars.

Q7. Design a basic Flutter application layout using layout widgets such as Row, Column, Container, Expanded, and Padding.

Code:

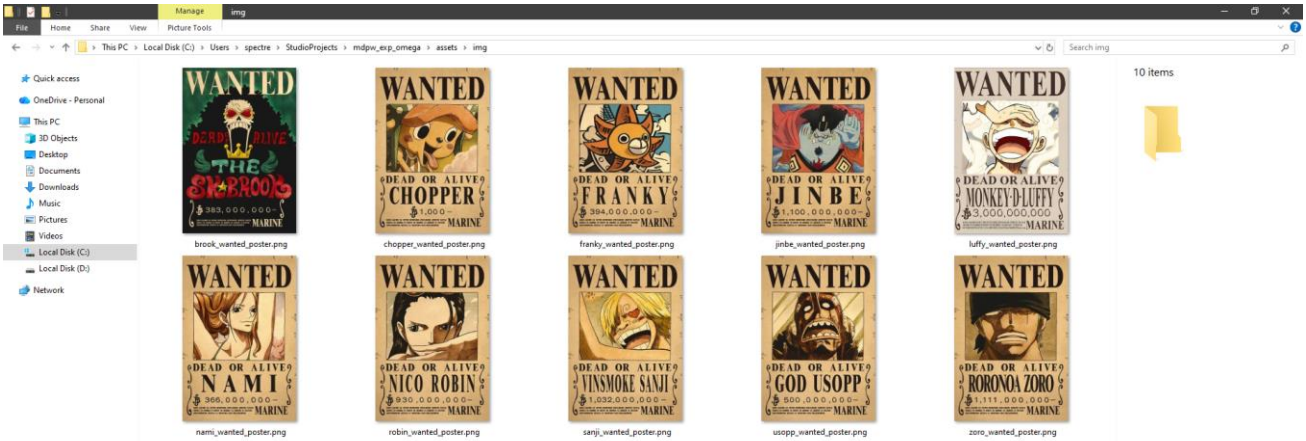
```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: HomeScreen(),
    );
  }
}

class StrawHat {
  final String fullName;
  final String alias;
  final String role;
  final String age;
  final String gender;
  final String description;
  final String image;

  StrawHat({
    required this.fullName,
    required this.alias,
    required this.role,
    required this.age,
    required this.gender,
    required this.description,
    required this.image,
  });
}

class HomeScreen extends StatelessWidget
{
  HomeScreen({super.key});

  final List<StrawHat> crew = [
    StrawHat(
      fullName: "Monkey D. Luffy",
      alias: "Straw Hat",
      role: "Captain",
      age: "19",
      gender: "Male",
      description: "Future King of the
Pirates with rubber powers.",
      image:
"assets/img/luffy_wanted_poster.png",
    ),
    StrawHat(
      fullName: "Roronoa Zoro",
      alias: "Pirate Hunter",
      role: "Swordsman",
      age: "21",
      gender: "Male",
      description: "Master of Three-Sword
Style.",
      image:
"assets/img/zoro_wanted_poster.png",
    ),
    StrawHat(
      fullName: "Nami",
      alias: "Cat Burglar",
      role: "Navigator",
      age: "20",
      gender: "Female",
      description: "Expert navigator and
weather specialist.",
      image:
```

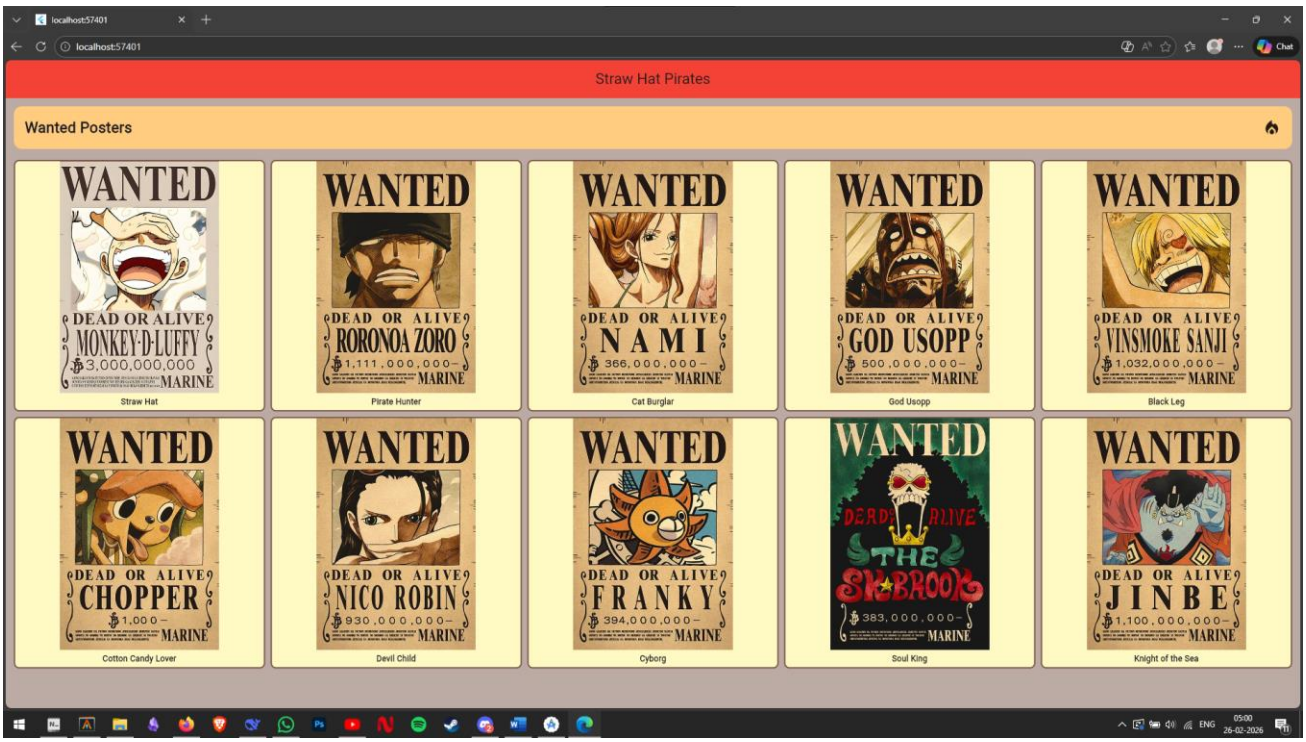
```
"assets/img/nami_wanted_poster.png",
    ),
    StrawHat(
      fullName: "Usopp",
      alias: "God Usopp",
      role: "Sniper",
      age: "19",
      gender: "Male",
      description: "Sharpshooter and
brave warrior of the sea.",
      image:
"assets/img/usopp_wanted_poster.png",
    ),
    StrawHat(
      fullName: "Sanji",
      alias: "Black Leg",
      role: "Cook",
      age: "21",
      gender: "Male",
      description: "Martial artist and
master chef.",
      image:
"assets/img/sanji_wanted_poster.png",
    ),
    StrawHat(
      fullName: "Tony Tony Chopper",
      alias: "Cotton Candy Lover",
      role: "Doctor",
      age: "17",
      gender: "Male",
      description: "Reindeer with human-
human fruit powers.",
      image:
"assets/img/chopper_wanted_poster.png",
    ),
    StrawHat(
      fullName: "Nico Robin",
      alias: "Devil Child",
      role: "Archaeologist",
      age: "30",
      gender: "Female",
      description: "Can sprout body parts
anywhere.",
      image:
"assets/img/robin_wanted_poster.png",
    ),
    StrawHat(
      fullName: "Franky",
      alias: "Cyborg",
      role: "Shipwright",
      age: "36",
      gender: "Male",
      description: "Super cyborg
shipbuilder.",
      image:
"assets/img/franky_wanted_poster.png",
    ),
    StrawHat(
      fullName: "Brook",
      alias: "Soul King",
      role: "Musician",
      age: "90",
      gender: "Male",
      description: "Living skeleton
musician.",
      image:
"assets/img/brook_wanted_poster.png",
    ),
    StrawHat(
      fullName: "Jinbe",
      alias: "Knight of the Sea",
      role: "Helmsman",
      age: "46",
      gender: "Male",
      description: "Former Warlord of the
Sea and master of Fish-Man Karate. A
loyal and honorable helmsman of the Straw
Hat Pirates.",
      image:
"assets/img/jinbe_wanted_poster.png",
    ),
  ];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: Colors.brown[200],
      appBar: AppBar(
        title: const Text("Straw Hat
Pirates"),
        centerTitle: true,
        backgroundColor: Colors.red,
      ),
      body: Padding(
        padding: const
EdgeInsets.all(12.0),
        child: Column(
          children: [
            // Header Section
            Container(
              padding: const
EdgeInsets.all(16),
              decoration: BoxDecoration(
                color:
Colors.orange[200],
                borderRadius:
BorderRadius.circular(12),
              ),
              child: const Row(
                children: [
                  Expanded(
                    child: Text(
                      "Wanted Posters",
                      style: TextStyle(
```

```
                        fontSize: 22,
                        fontWeight:
FontWeight.bold,
                      ),
                    ),
                  ),

Icon(Icons.local_fire_department, size:
28),
                ],
              ),
            ),
            const SizedBox(height: 16),

            // Grid Section
            Expanded(
              child: GridView.builder(
                itemCount: crew.length,
                gridDelegate:
                const
SliverGridDelegateWithFixedCrossAxisCount
(
                  crossAxisCount: 5, // 5
columns
                  crossAxisSpacing: 8,
                  mainAxisSpacing: 8,
                ),
                itemBuilder: (context,
index) {
                  final member =
crew[index];
                  return GestureDetector(
                    onTap: () =>
showDetails(context, member),
                    child: Container(
                      decoration:
BoxDecoration(
                        color:
Colors.yellow[100],
                        borderRadius:
BorderRadius.circular(8),
                        border:
Border.all(color: Colors.brown, width:
2),
                      ),
                      child: Column(
                        children: [
                          Expanded(
                            child:
Image.asset(

member.image,
                              fit:
BoxFit.cover,
                            ),
                          ),
                          Padding(
                            padding:
const EdgeInsets.all(4.0),
                            child: Text(

member.alias,
                              style:
const TextStyle(
                                fontSize:
12,

fontWeight: FontWeight.bold,
                              ),
                              textAlign:
TextAlign.center,
                            ),
                          ),
                        ],
                      ),
                    ),
                  );
                },
              ),
            ),
          ),
        ),
      );
    }

    void showDetails(BuildContext context,
StrawHat member) {
      showModalBottomSheet(
        context: context,
        isScrollControlled: true,
        builder: (context) {
          return Padding(
            padding: const
EdgeInsets.all(16.0),
            child: Column(
              mainAxisSize:
MainAxisSize.min,
              crossAxisAlignment:
CrossAxisAlignment.start,
              children: [
                Text(member.fullName,
                    style: const TextStyle(
                        fontSize: 20,
fontWeight: FontWeight.bold)),
                const SizedBox(height: 10),
                Text("Alias:
${member.alias}"),
                Text("Role:
${member.role}"),
                Text("Age: ${member.age}"),
                Text("Gender:
${member.gender}"),
                const SizedBox(height: 10),
```

```
            Text("Description:                          );
${member.description}"),                              },
            const SizedBox(height: 20),             );
          ],                                      }
        ),                                      }
```

Output:





Conclusion

This experiment successfully demonstrated Flutter's core layout widgets—Row, Column, Container, Expanded, and Padding. These widgets were effectively combined to create a responsive, well-structured interface.